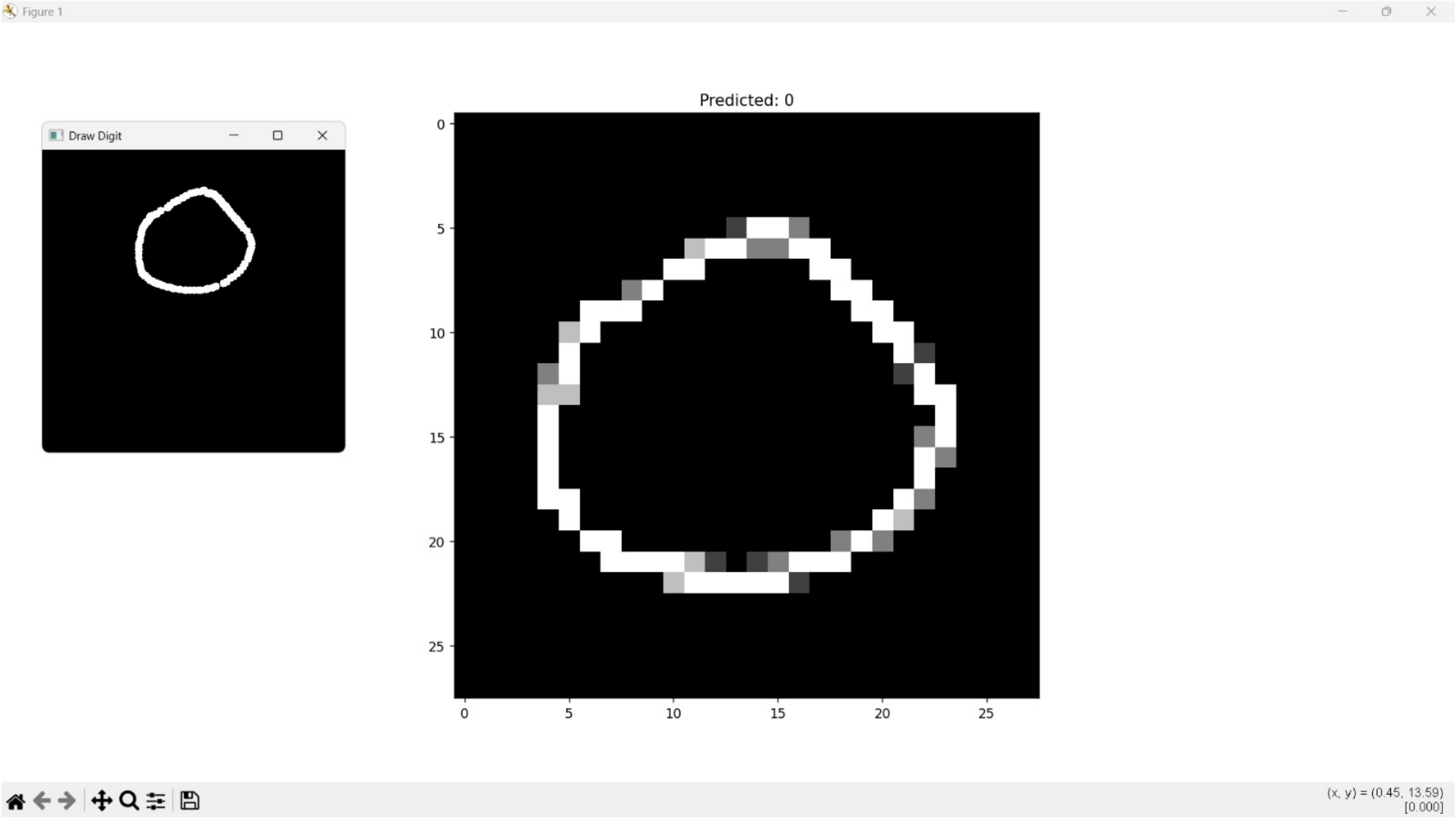


Figure 1
Draw Digit
Predicted: 0
0
5
10
15
20
25
0
5
10
15
20
25
(x, y) = (0.45, 13.59)
[0.000]

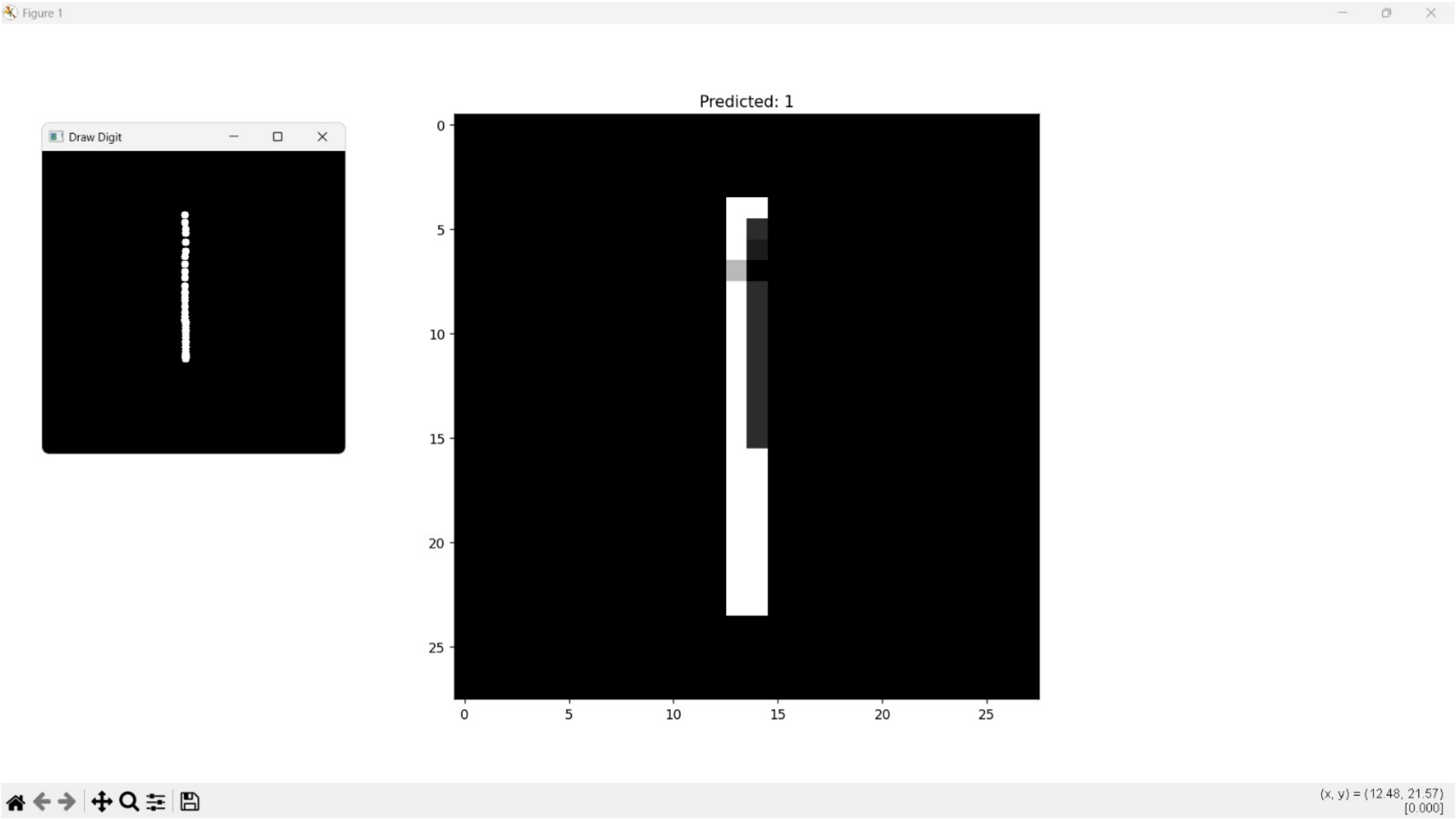


Figure 1
Draw Digit
Predicted: 1
0
5
10
15
20
25
0
5
10
15
20
25
(x, y) = (12.48, 21.57)
[0.000]

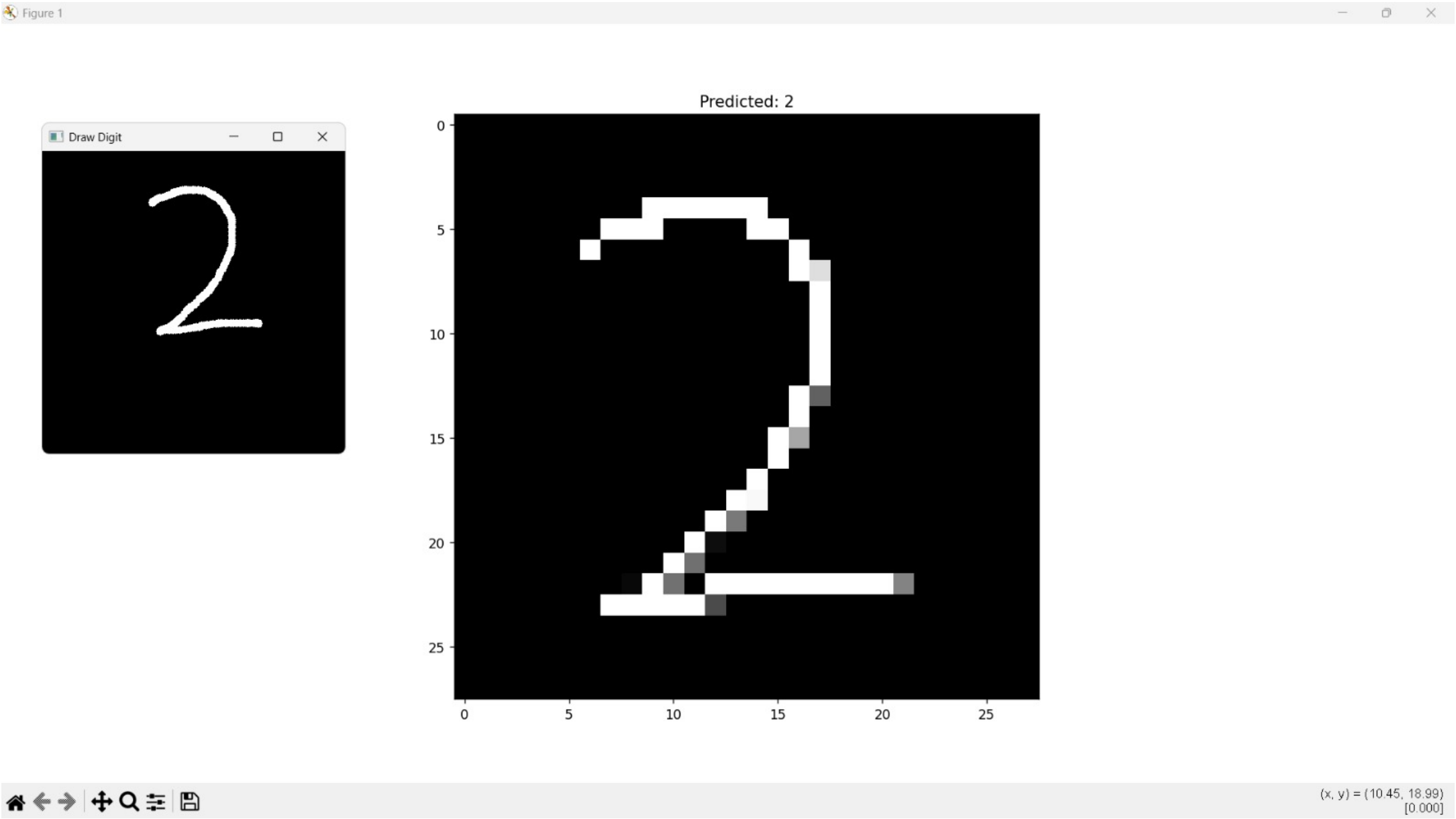


Figure 1
Draw Digit
Predicted: 2
0
5
10
15
20
25
0
5
10
15
20
25
(x, y) = (10.45, 18.99)
[0.000]

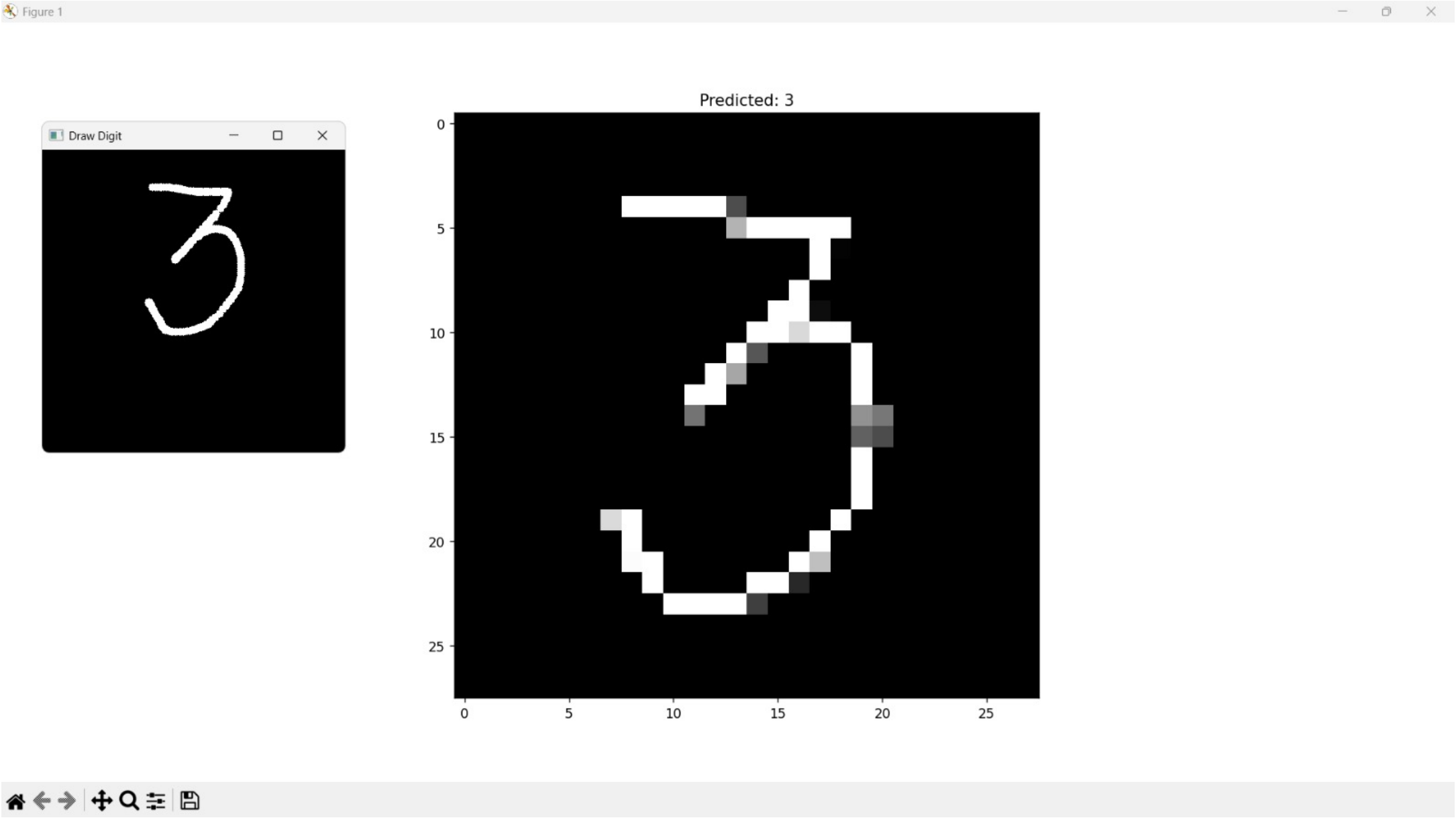


Figure 1
Draw Digit
Predicted: 3
0
5
10
15
20
25
0
5
10
15
20
25

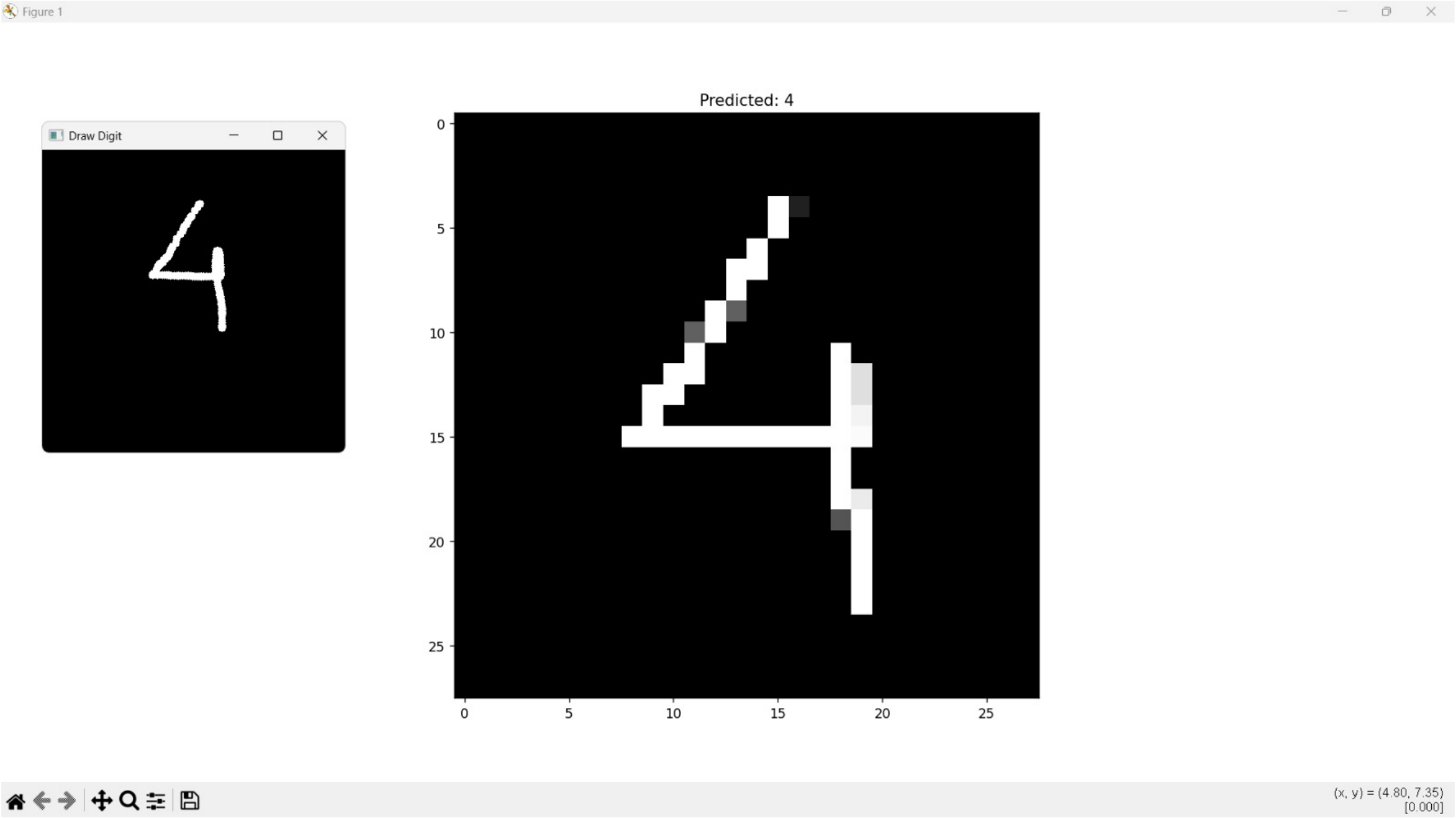


Figure 1
Draw Digit
Predicted: 4
0
5
10
15
20
25
0
5
10
15
20
25
(x, y) = (4.80, 7.35)
[0.000]

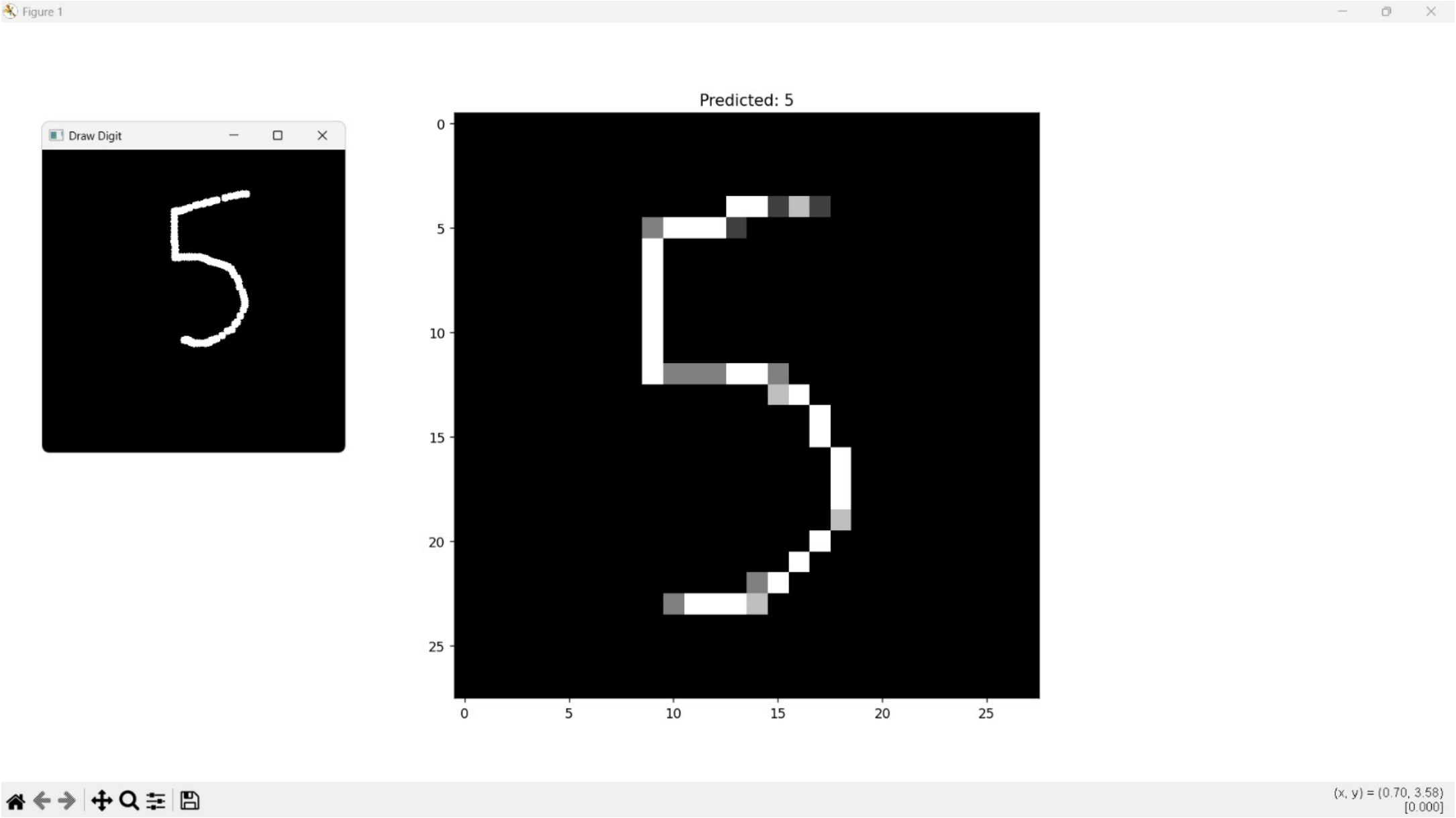


Figure 1
Draw Digit
Predicted: 5
0
5
10
15
20
25
0
5
10
15
20
25
(x, y) = (0.70, 3.58)
[0.000]

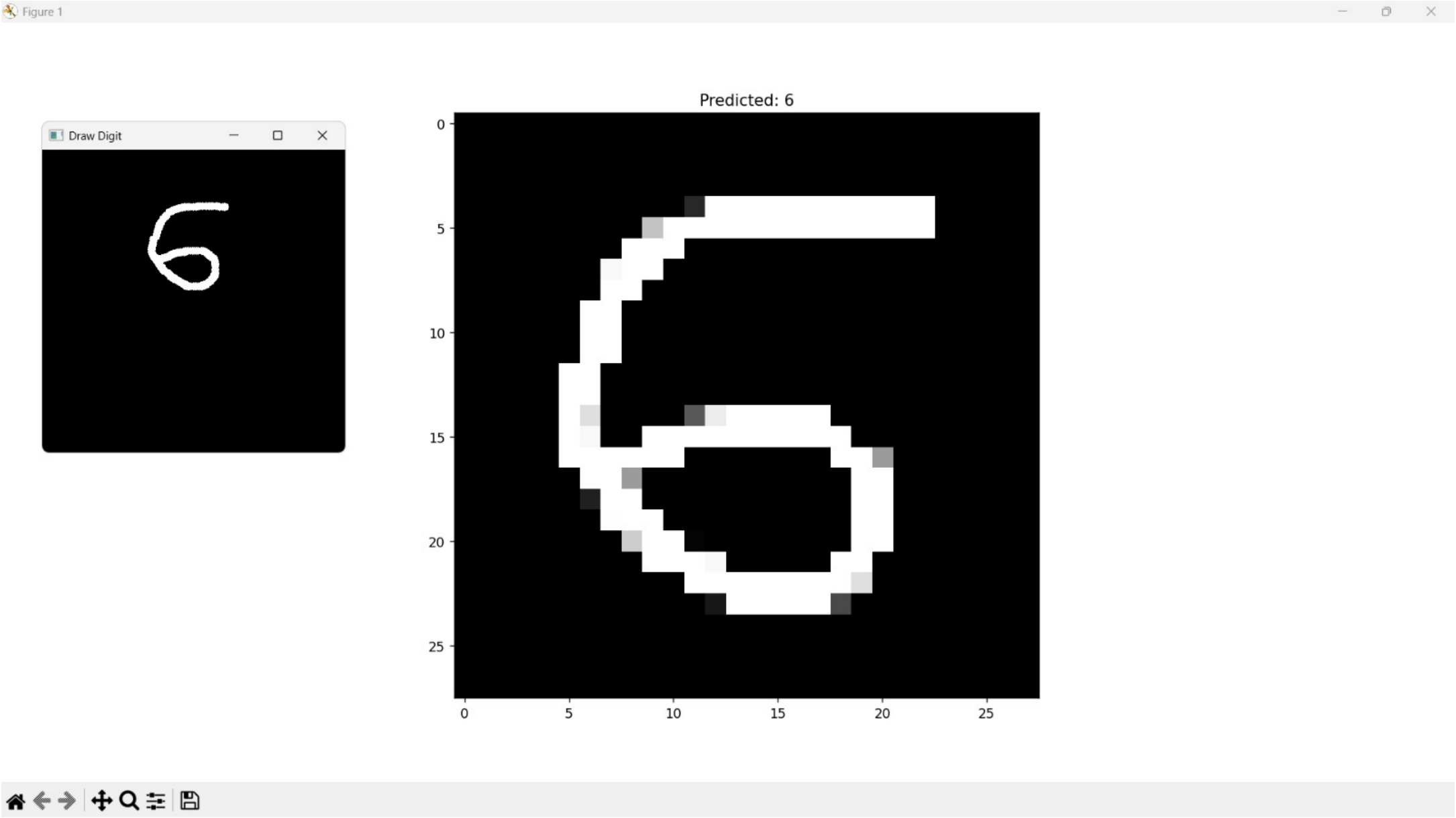


Figure 1
Draw Digit
Predicted: 6
0
5
10
15
20
25
0
5
10
15
20
25

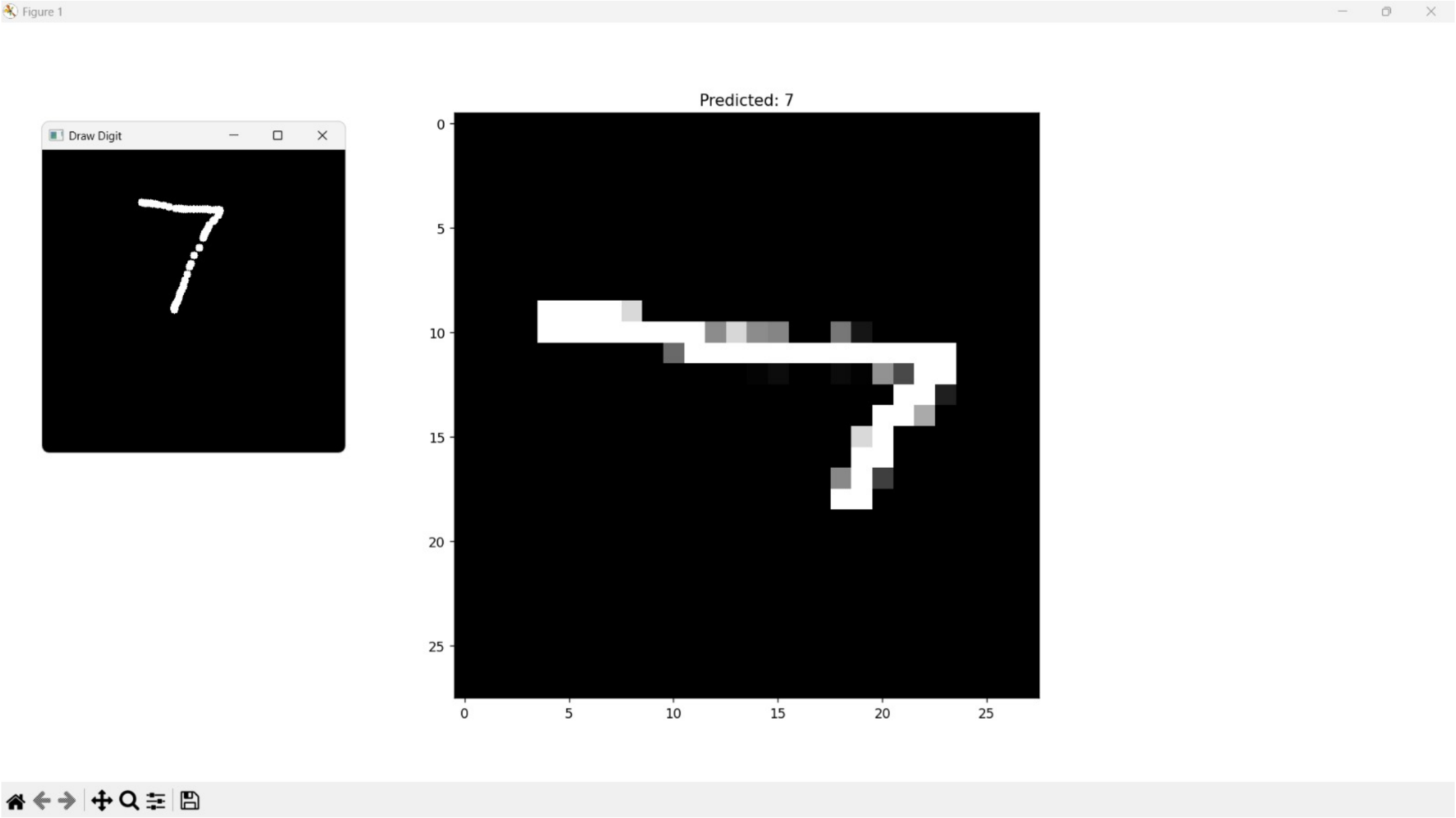


Figure 1
Draw Digit
Predicted: 7
0
5
10
15
20
25
0
5
10
15
20
25

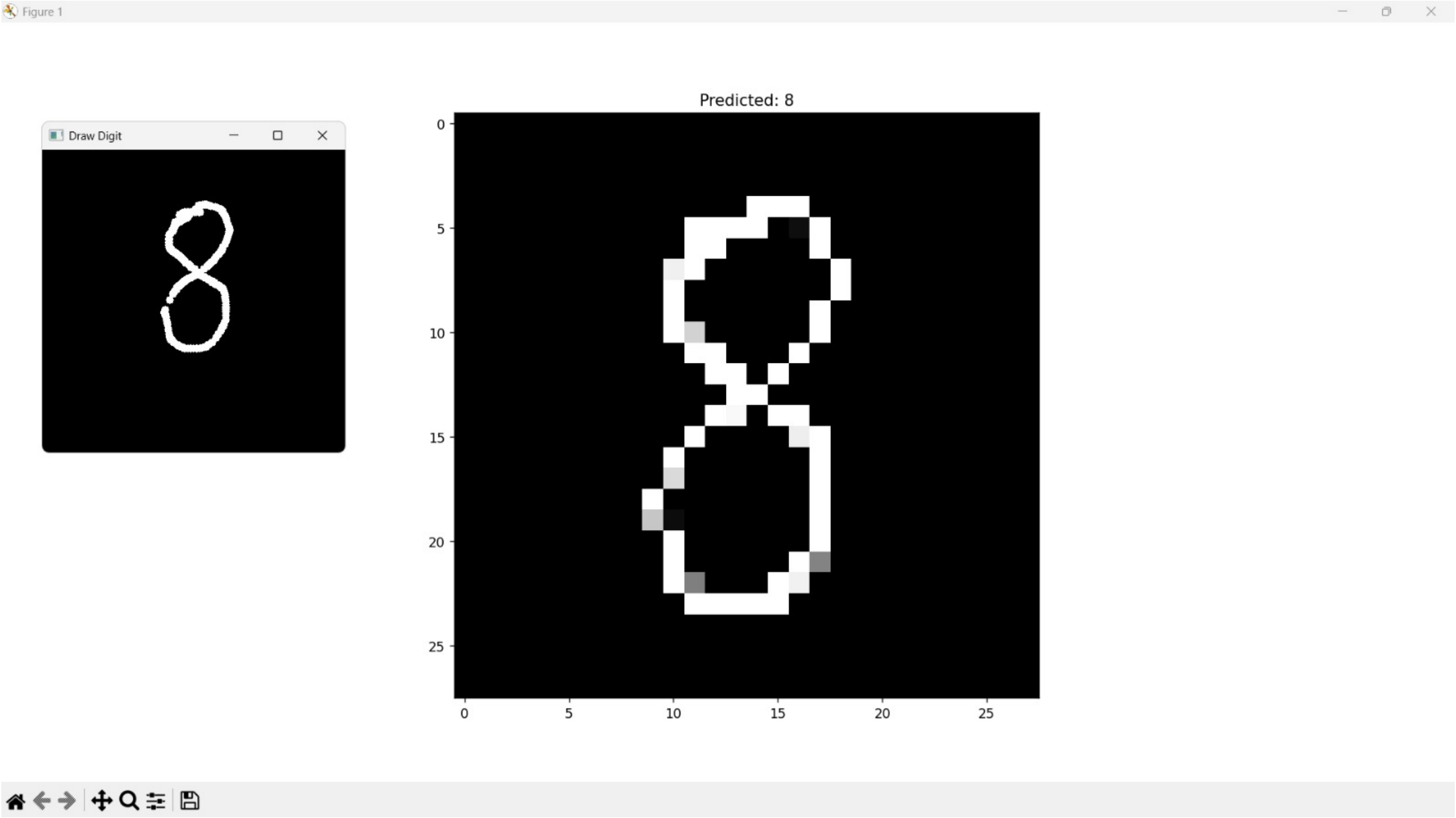


Figure 1
Draw Digit
Predicted: 8
0
5
10
15
20
25
0
5
10
15
20
25

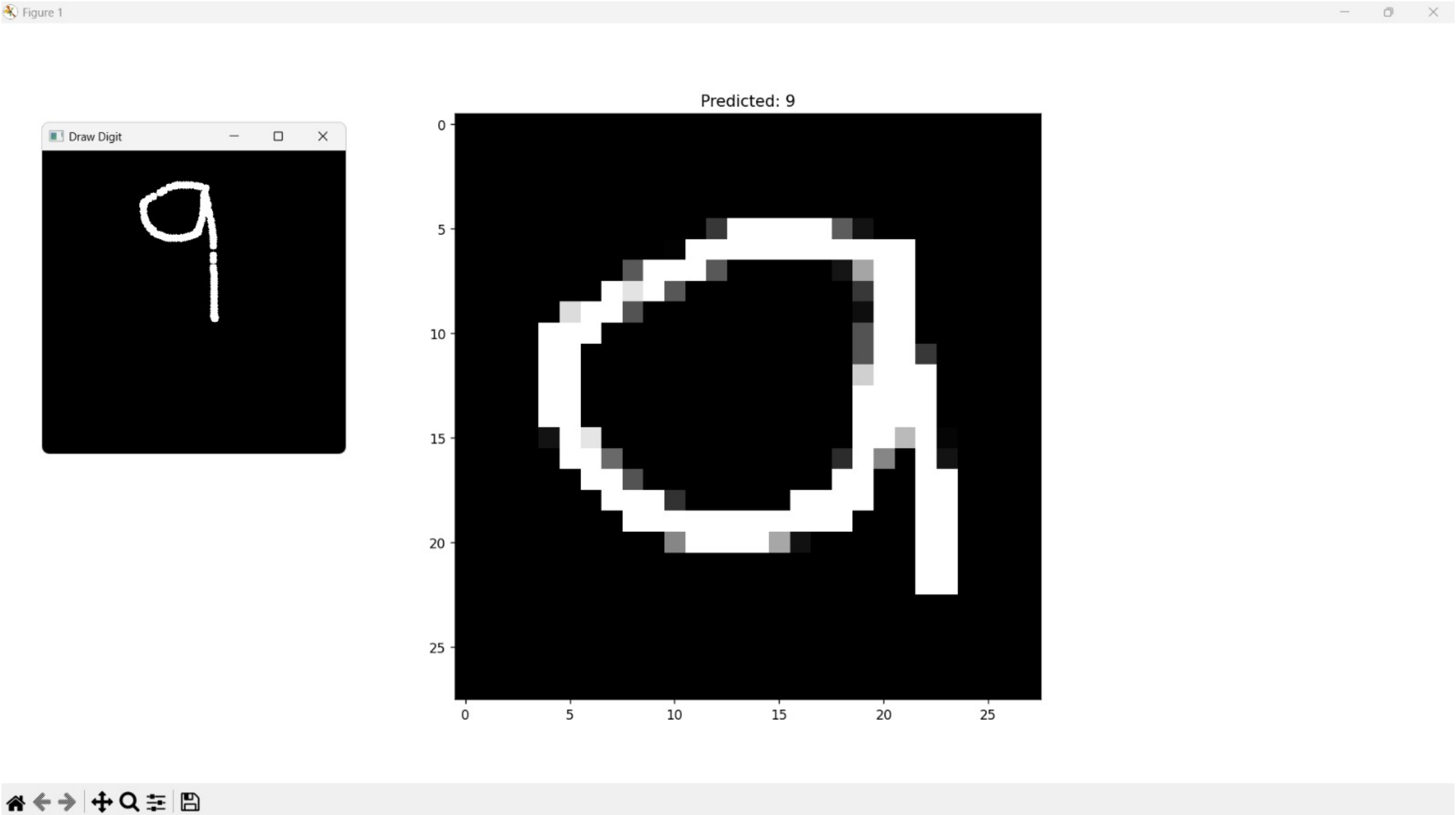
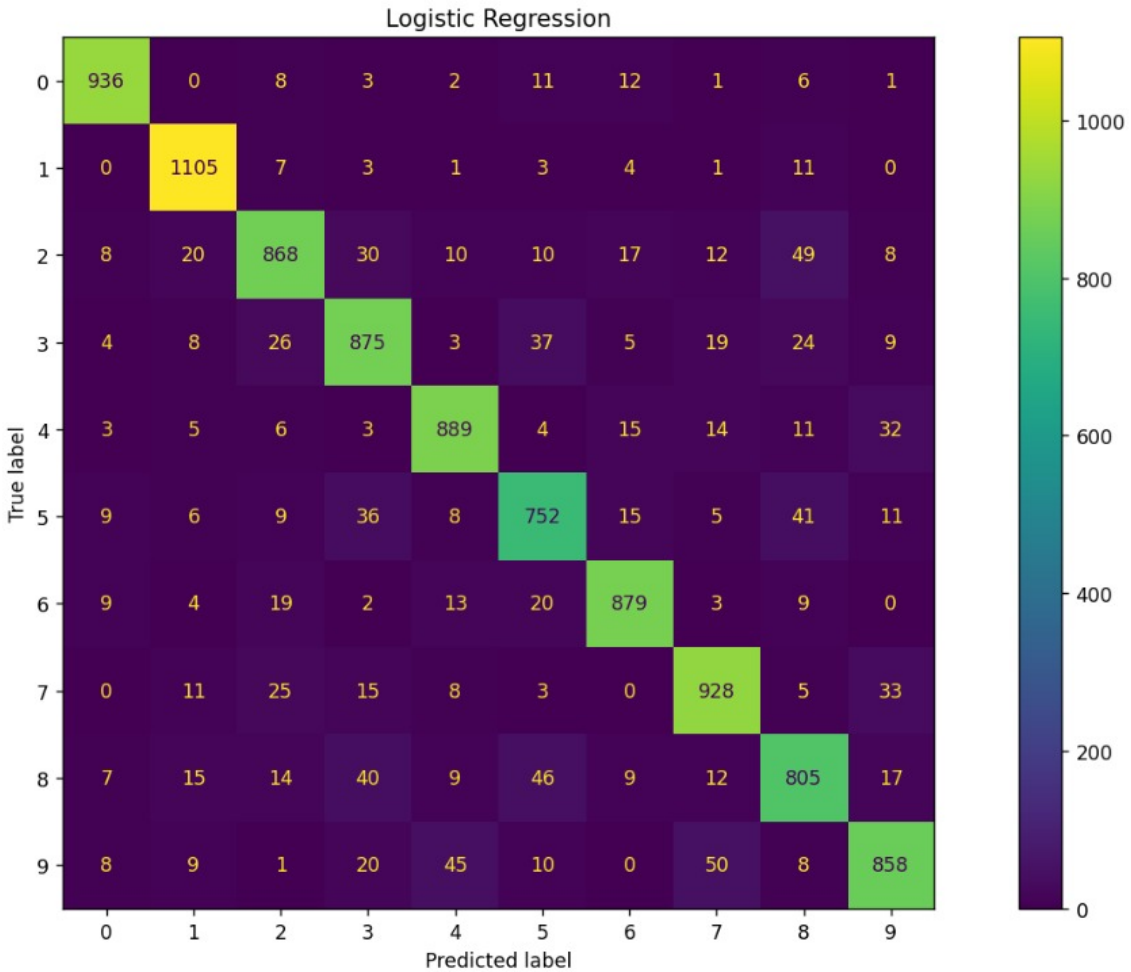


Figure 1
Draw Digit
Predicted: 9
0
5
10
15
20
25
0
5
10
15
20
25

Figure 1

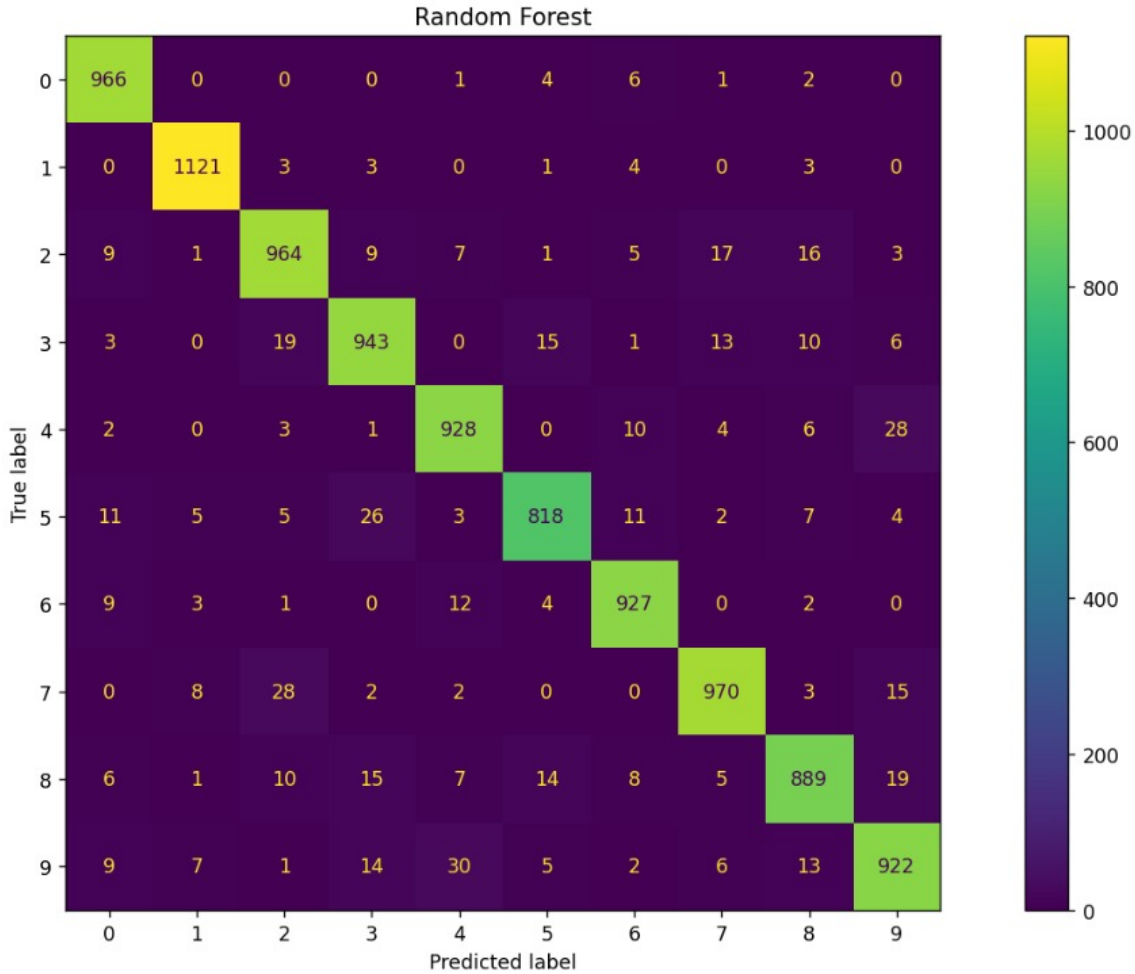

Logistic Regression
True label
Predicted label
0 936 0 8 3 2 11 12 1 6 1
1 0 1105 7 3 1 3 4 1 11 0
2 8 20 868 30 10 10 17 12 49 8
3 4 8 26 875 3 37 5 19 24 9
4 3 5 6 3 889 4 15 14 11 32
5 9 6 9 36 8 752 15 5 41 11
6 9 4 19 2 13 20 879 3 9 0
7 0 11 25 15 8 3 0 928 5 33
8 7 15 14 40 9 46 9 12 805 17
9 8 9 1 20 45 10 0 50 8 858
0 1 2 3 4 5 6 7 8 9
1000
800
600
400
200
0

```
Logistic Regression
Accuracy: 0.8895
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.96      0.95       980
           1       0.93      0.97      0.95      1135
           2       0.88      0.84      0.86      1032
           3       0.85      0.87      0.86      1010
           4       0.90      0.91      0.90       982
           5       0.84      0.84      0.84       892
           6       0.92      0.92      0.92       958
           7       0.89      0.90      0.90      1028
           8       0.83      0.83      0.83       974
           9       0.89      0.85      0.87      1009

    accuracy                           0.89     10000
   macro avg       0.89      0.89      0.89     10000
weighted avg       0.89      0.89      0.89     10000
```

Figure 1


Random Forest
True label
Predicted label
0 1 2 3 4 5 6 7 8 9
966 0 0 0 1 4 6 1 2 0
0 1121 3 3 0 1 4 0 3 0
9 1 964 9 7 1 5 17 16 3
3 0 19 943 0 15 1 13 10 6
2 0 3 1 928 0 10 4 6 28
11 5 5 26 3 818 11 2 7 4
9 3 1 0 12 4 927 0 2 0
0 8 28 2 2 0 0 970 3 15
6 1 10 15 7 14 8 5 889 19
9 7 1 14 30 5 2 6 13 922
0 200 400 600 800 1000

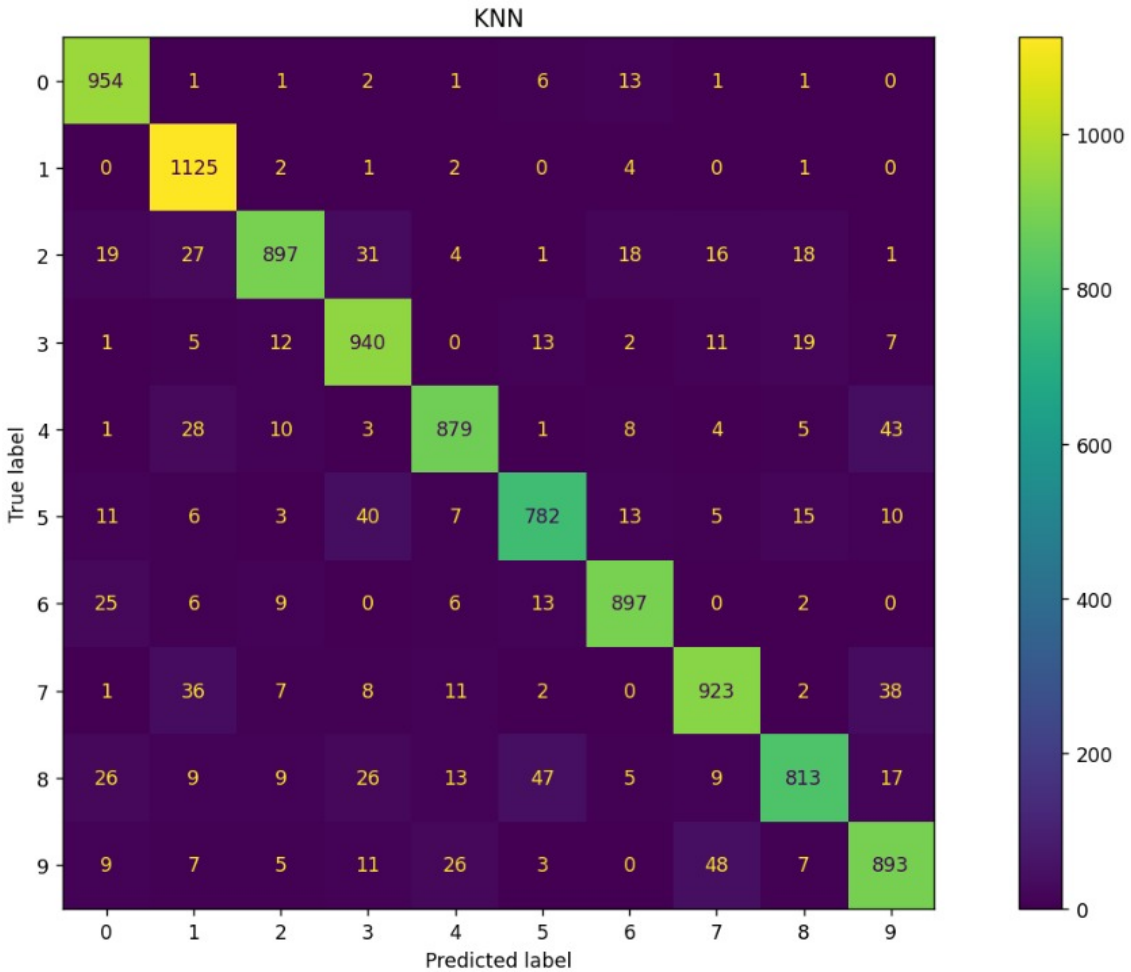
(x, y) = (,)
[1.]

```
Random Forest
Accuracy: 0.9448
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.99      0.97       980
           1       0.98      0.99      0.98      1135
           2       0.93      0.93      0.93      1032
           3       0.93      0.93      0.93      1010
           4       0.94      0.95      0.94       982
           5       0.95      0.92      0.93       892
           6       0.95      0.97      0.96       958
           7       0.95      0.94      0.95      1028
           8       0.93      0.91      0.92       974
           9       0.92      0.91      0.92      1009

    accuracy                           0.94     10000
   macro avg       0.94      0.94      0.94     10000
weighted avg       0.94      0.94      0.94     10000
```

Figure 1

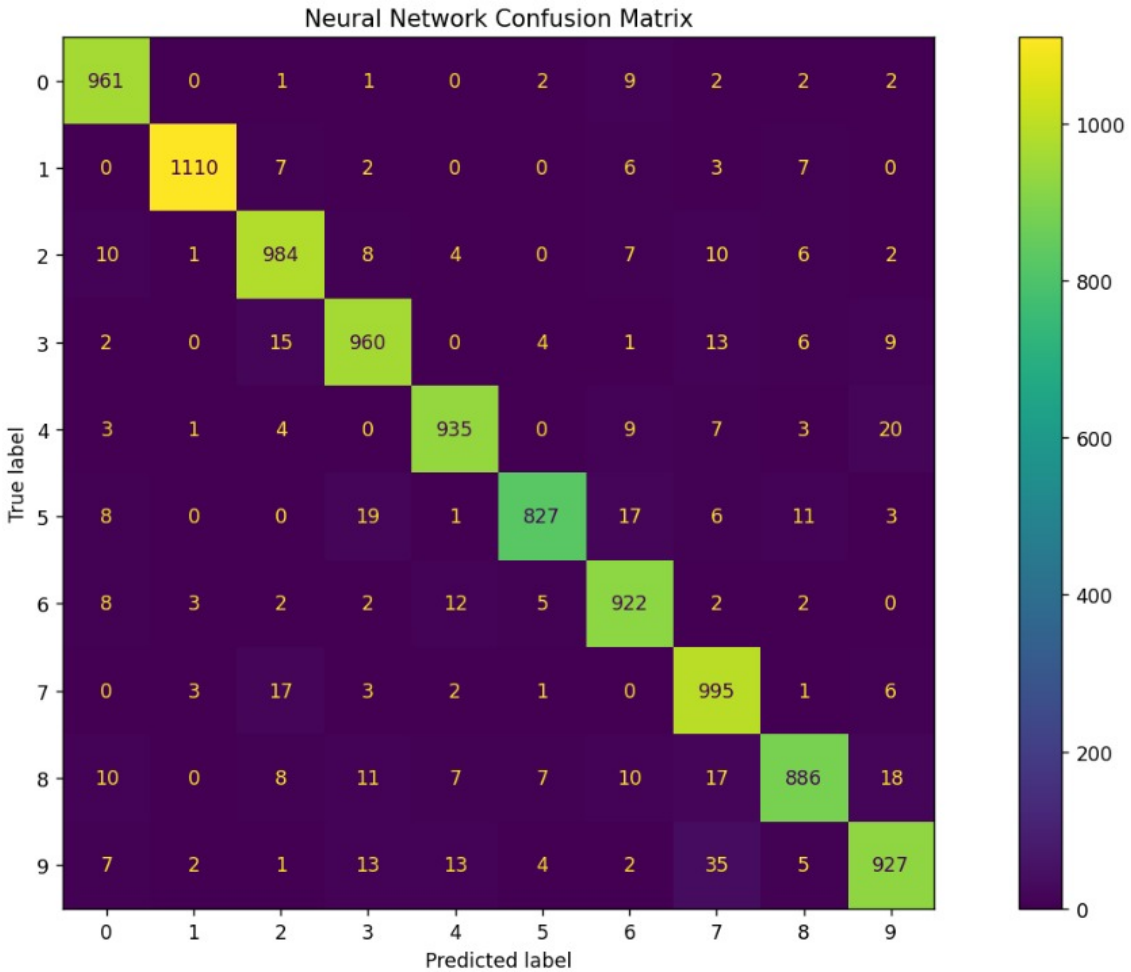

KNN
True label
Predicted label
0 954 1 1 2 1 6 13 1 1 0
1 0 1125 2 1 2 0 4 0 1 0
2 19 27 897 31 4 1 18 16 18 1
3 1 5 12 940 0 13 2 11 19 7
4 1 28 10 3 879 1 8 4 5 43
5 11 6 3 40 7 782 13 5 15 10
6 25 6 9 0 6 13 897 0 2 0
7 1 36 7 8 11 2 0 923 2 38
8 26 9 9 26 13 47 5 9 813 17
9 9 7 5 11 26 3 0 48 7 893
0 1 2 3 4 5 6 7 8 9
0 200 400 600 800 1000

```
KNN
Accuracy: 0.9103
Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.97      0.94       980
           1       0.90      0.99      0.94      1135
           2       0.94      0.87      0.90      1032
           3       0.89      0.93      0.91      1010
           4       0.93      0.90      0.91       982
           5       0.90      0.88      0.89       892
           6       0.93      0.94      0.94       958
           7       0.91      0.90      0.90      1028
           8       0.92      0.83      0.88       974
           9       0.89      0.89      0.89      1009

    accuracy                           0.91     10000
   macro avg       0.91      0.91      0.91     10000
weighted avg       0.91      0.91      0.91     10000
```

Figure 1


Neural Network Confusion Matrix
True label
Predicted label
0 1 2 3 4 5 6 7 8 9
961 0 1 1 0 2 9 2 2 2
0 1110 7 2 0 0 6 3 7 0
10 1 984 8 4 0 7 10 6 2
2 0 15 960 0 4 1 13 6 9
3 1 4 0 935 0 9 7 3 20
8 0 0 19 1 827 17 6 11 3
8 3 2 2 12 5 922 2 2 0
0 3 17 3 2 1 0 995 1 6
10 0 8 11 7 7 10 17 886 18
7 2 1 13 13 4 2 35 5 927
0 200 400 600 800 1000

(x, y) = (,)
[1.]

```
===== NEURAL NETWORK =====
C:\Users\lenovo\AppData\Roaming\Python\Python312\site-packages\keras\src\layers\core\dense.py:95: UserWarning: Do not pass an `input_shape`/`input_dim`
 argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
2026-04-01 23:53:53.084867: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions
in performance-critical operations.
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
Epoch 1/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.8441 - loss: 0.5540 - val_accuracy: 0.9180 - val_loss: 0.2859
Epoch 2/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9362 - loss: 0.2277 - val_accuracy: 0.9370 - val_loss: 0.2080
Epoch 3/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9530 - loss: 0.1604 - val_accuracy: 0.9440 - val_loss: 0.1998
Epoch 4/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9681 - loss: 0.1108 - val_accuracy: 0.9460 - val_loss: 0.1845
Epoch 5/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9744 - loss: 0.0847 - val_accuracy: 0.9530 - val_loss: 0.1447
Epoch 6/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9838 - loss: 0.0599 - val_accuracy: 0.9530 - val_loss: 0.1550
Epoch 7/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9857 - loss: 0.0464 - val_accuracy: 0.9510 - val_loss: 0.1796
Epoch 8/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9927 - loss: 0.0288 - val_accuracy: 0.9470 - val_loss: 0.1767
Epoch 9/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9939 - loss: 0.0236 - val_accuracy: 0.9470 - val_loss: 0.1898
Epoch 10/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9970 - loss: 0.0162 - val_accuracy: 0.9540 - val_loss: 0.1922
313/313 ━━━━━━━━━━━━━━━━━━━━ 0s 995us/step - accuracy: 0.9507 - loss: 0.1779
Neural Network Accuracy: 0.9506999850273132
313/313 ━━━━━━━━━━━━━━━━━━━━ 0s 664us/step
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.98      0.97       980
           1       0.99      0.98      0.98      1135
           2       0.95      0.95      0.95      1032
           3       0.94      0.95      0.95      1010
           4       0.96      0.95      0.96       982
           5       0.97      0.93      0.95       892
           6       0.94      0.96      0.95       958
           7       0.91      0.97      0.94      1028
           8       0.95      0.91      0.93       974
           9       0.94      0.92      0.93      1009

    accuracy                           0.95     10000
   macro avg       0.95      0.95      0.95     10000
weighted avg       0.95      0.95      0.95     10000
```

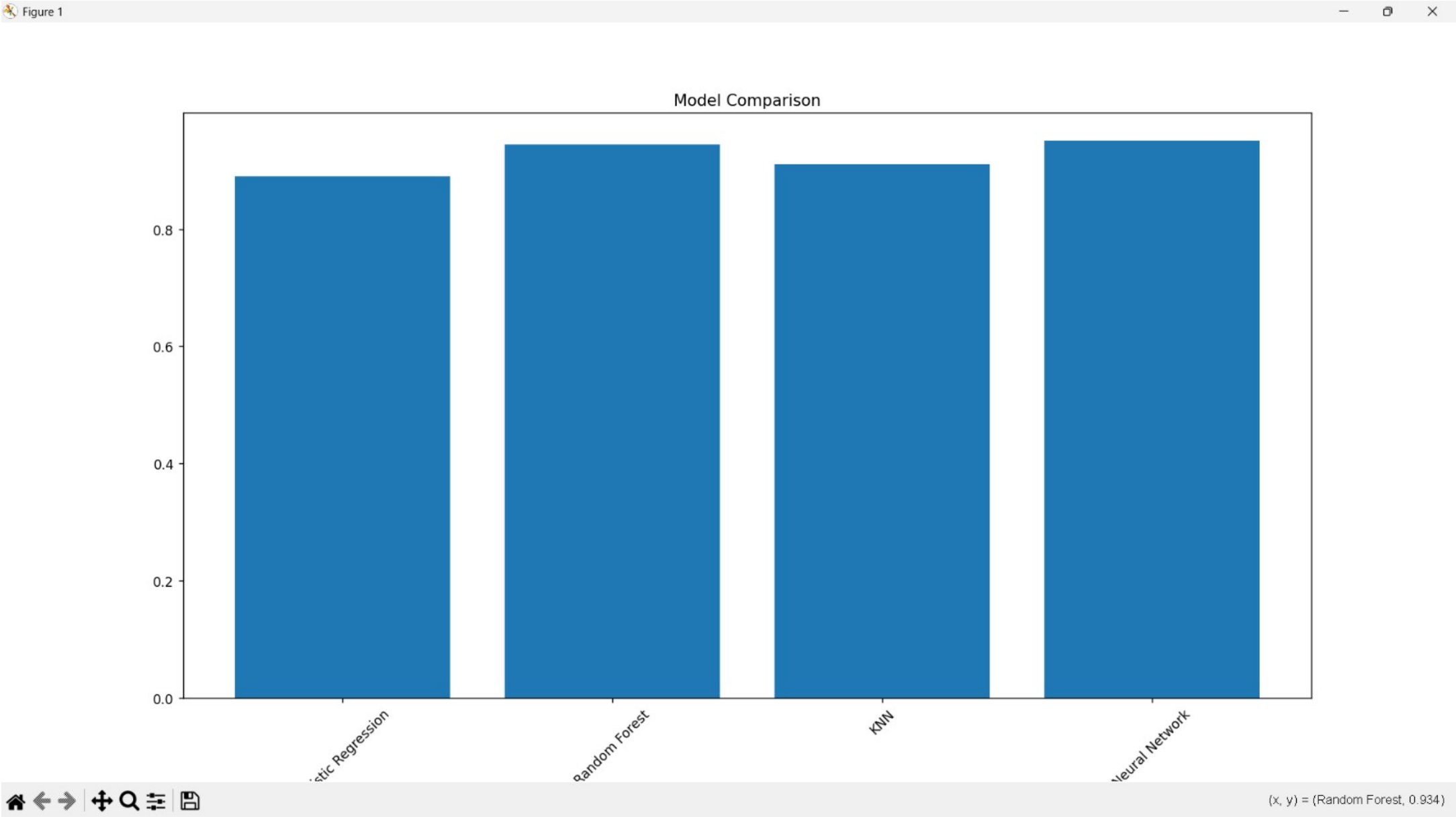


Figure 1
Model Comparison
0.8
0.6
0.4
0.2
0.0
KNN
(x, y) = (Random Forest, 0.934)

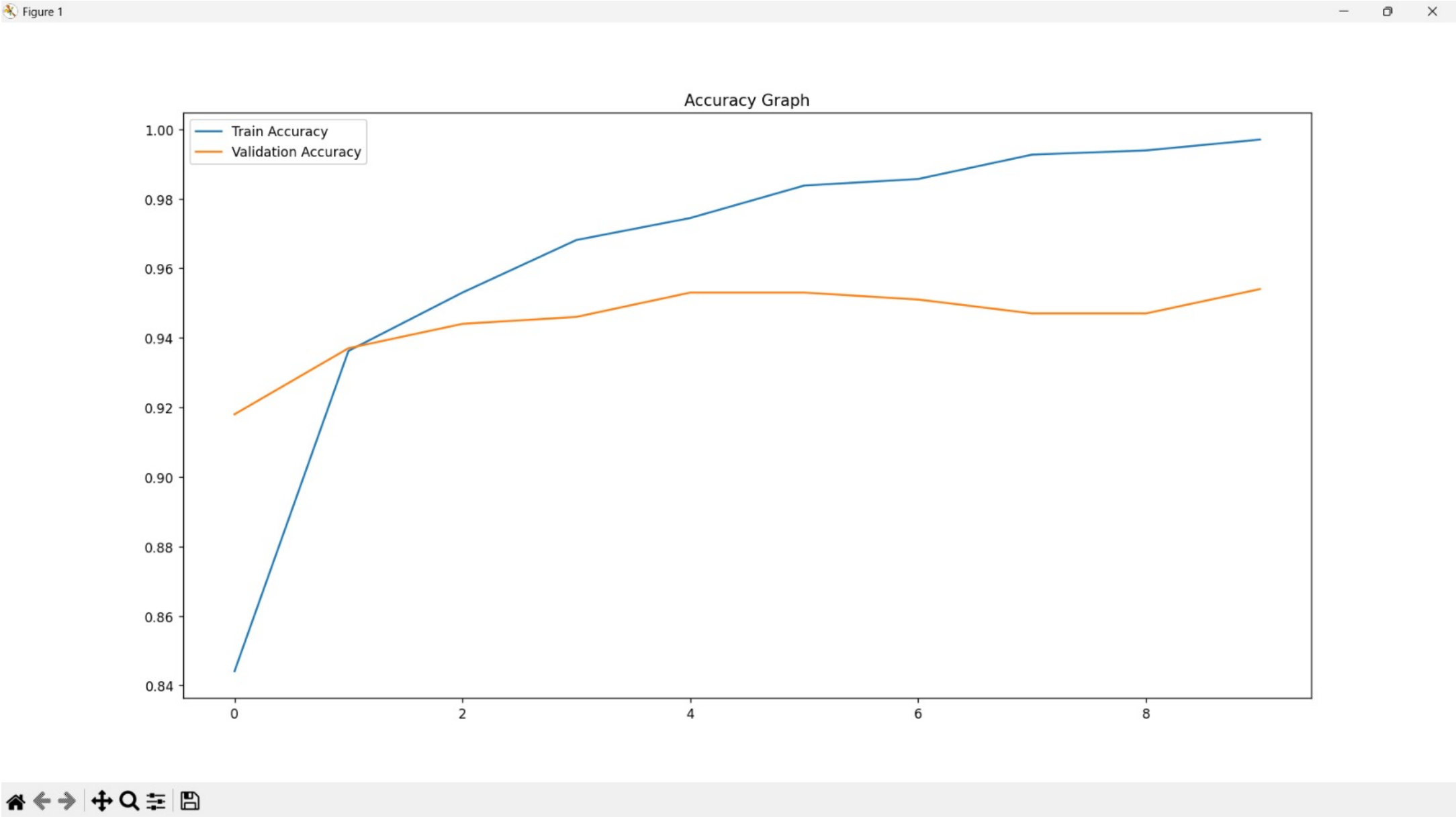


Figure 1
Accuracy Graph
Train Accuracy
Validation Accuracy
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.84
0
2
4
6
8

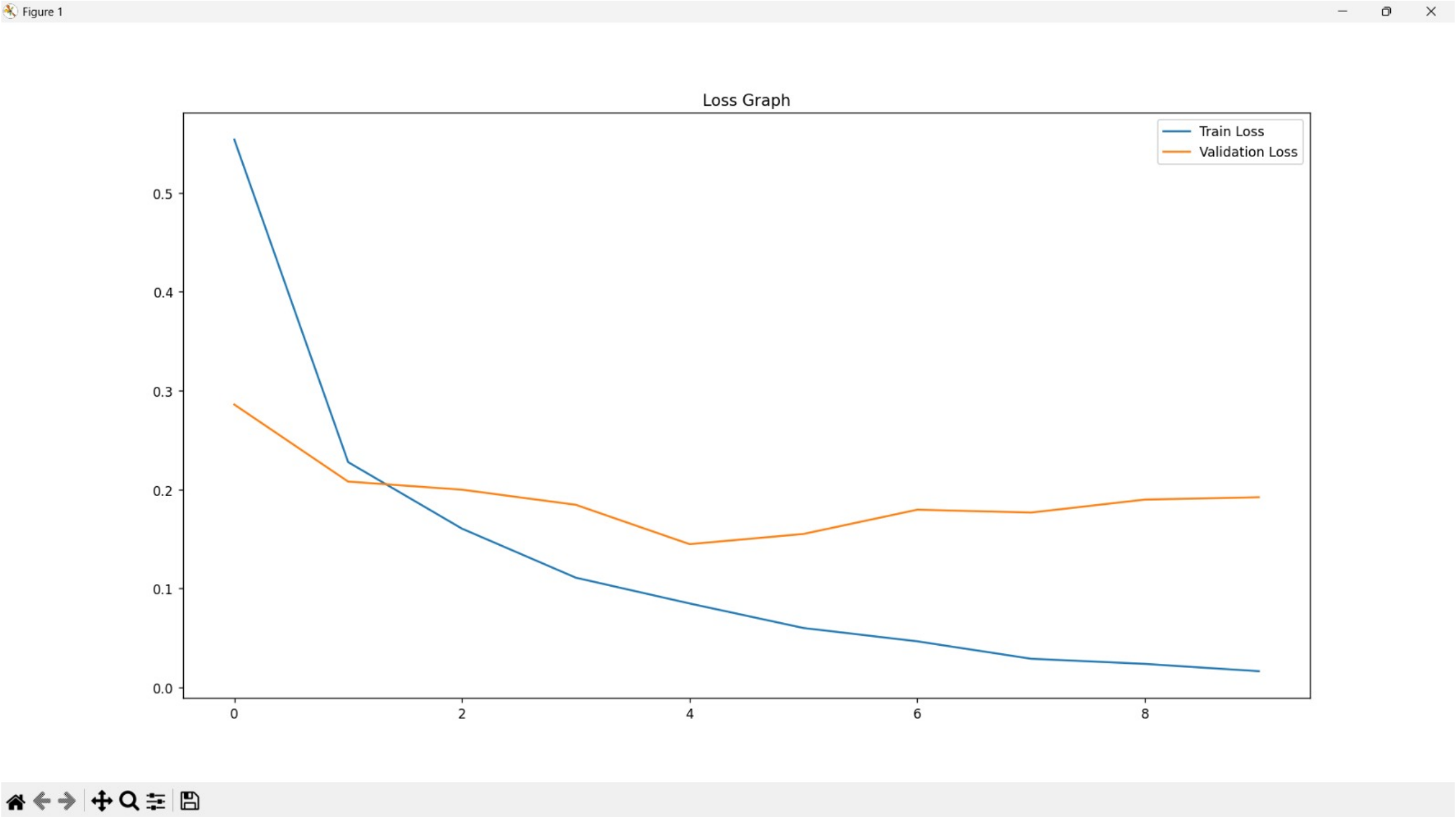


Figure 1
Loss Graph
Train Loss
Validation Loss
0.5
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8